ELECTRICAL MACHINES · MOCK

Machines mock paper

Answers and worked solutions · Day 23 · Sun 15 Nov · pack MX3

Marking guide

All numbers below were computed in Python and are given to 4 significant figures. Accept answers within about 0.5 % that come from sensible intermediate rounding.

Question 1 marking guide

(i) KVL around the loop:

$$V_t = E - IR_{cable} = 48 - 40(0.12) = \mathbf{43.2\ V}$$

$$P_{cable} = I^2 R_{cable} = 40^2(0.12) = \mathbf{192\ W}$$

(ii)

$$P_{in,motor} = V_t I = 43.2 \times 40 = \mathbf{1728\ W}$$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{1500}{1728} = \mathbf{86.81\ \%}$$

$$\omega = \frac{2\pi \times 1200}{60} = 125.7\ \text{rad/s}, \qquad \tau = \frac{P_{out}}{\omega} = \frac{1500}{125.7} = \mathbf{11.94\ Nm}$$

(Check: the source supplies $48 \times 40 = 1920$ W $= 1728 + 192$.)

MARKS: (i) 1 for V_t, 1 for the cable loss. (ii) 1 each for input power, efficiency and torque.

> COMMON MISTAKES:

- Using $EI = 1920$ W as the motor input. That is the source output. The motor only receives $V_t I$. (The overall efficiency from the source, 78.13 %, is not what was asked.)
- Using rpm instead of rad/s in $\tau = P/\omega$.

Method from Lecture 2.8.3 (single-phase analysis, delta): convert to phase quantities, analyse one phase, convert back.

(i) In delta the phase voltage equals the line voltage: $\mathbf{V}_1 = 415\angle 0^\circ$ V.

$$Z = 24 + j18 = 30\angle 36.87^\circ\ \Omega$$

$$\mathbf{I}_1 = \frac{\mathbf{V}_1}{Z} = \frac{415\angle 0^\circ}{30\angle 36.87^\circ} = \mathbf{13.83\angle -36.87^\circ\ A}$$

$$I_L = \sqrt{3} I_1 = \sqrt{3} \times 13.83 = \mathbf{23.96\ A}$$

$$\text{pf} = \cos 36.87^\circ = \mathbf{0.8\ lagging}$$

(ii)

$$P = 3I_1^2 R = 3(13.83)^2(24) = \mathbf{13.78\ kW}, \qquad Q = 3I_1^2 X = 3(13.83)^2(18) = \mathbf{10.33\ kVAr}$$

(Check: $P = \sqrt{3} V_L I_L \cos\phi = \sqrt{3}(415)(23.96)(0.8) = 13.78$ kW.)

(iii) In star each impedance sees $415/\sqrt{3} = 239.6$ V, so $I_1 = 239.6/30 = 7.987$ A and

$$P_{star} = 3(7.987)^2(24) = \mathbf{4.593\ kW}$$

This is one-third of the delta value, because the phase voltage drops by $\sqrt{3}$ and power goes as V^2.

MARKS: (i) 1 each for $\mathbf{I}_1$, I_L and pf. (ii) 1 each for P and Q. (iii) 1.

> **COMMON MISTAKES:**

- Dividing 415 V by $\sqrt{3}$ for a delta load. In delta, $V_1 = V_L$ and it is the current that has the $\sqrt{3}$.
- Reporting $I_L = I_1$ in delta.

(i) Motor input. The 18 kW is the shaft output, so

$$P_m = \frac{18}{0.9} = 20\ \text{kW}, \qquad Q_m = P_m \tan(\cos^{-1} 0.75) = 20 \times 0.8819 = 17.64\ \text{kVAr}$$

Add the delta load from 1b (powers add, currents do not):

$$P_T = 13.78 + 20 = \mathbf{33.78\ kW}, \qquad Q_T = 10.33 + 17.64 = \mathbf{27.97\ kVAr}$$

$$S_T = \sqrt{P_T^2 + Q_T^2} = 43.86\ \text{kVA}, \qquad \text{pf} = \frac{33.78}{43.86} = \mathbf{0.7702\ lagging}$$

$$I_L = \frac{S_T}{\sqrt{3} V_L} = \frac{43\,856}{\sqrt{3} \times 415} = \mathbf{61.01\ A}$$

(ii) The capacitors do not change P. The target supply reactive power is

$$Q_S = P_T \tan(\cos^{-1} 0.95) = 33.78 \times 0.3287 = 11.10\ \text{kVAr}$$

$$Q_C = Q_S - Q_T = 11.10 - 27.97 = \mathbf{-16.87\ kVAr}$$

(negative: the capacitors generate reactive power). The new apparent power is $S = 33.78/0.95 = 35.56$ kVA, so

$$I_L = \frac{35\,556}{\sqrt{3} \times 415} = \mathbf{49.47\ A}$$

The line current falls by about 19 %.

(iii) Each delta capacitor sees the line voltage, 415 V, and absorbs one-third of Q_C:

$$|Q_{C,ph}| = \frac{16.87}{3} = 5.623\ \text{kVAr} = \frac{V^2}{X_C} = V^2 \omega C$$

$$C = \frac{5623}{415^2 \times 2\pi \times 50} = \mathbf{103.9\ \mu F\ \text{per capacitor}}$$

(iv) For 0.95 leading, $Q_S = -11.10$ kVAr, so $Q_C = -11.10 - 27.97 = \mathbf{-39.07\ kVAr}$ (2.3 times as much). It is not done because the larger capacitors cost more for no reduction in line current (the line current is the same 49.47 A at 0.95 lead as at 0.95 lag), and a leading load can raise the supply voltage.

MARKS: (i) 1 for the motor input P and Q (dividing by efficiency), 1 for the totals, 1 for pf and I_L. (ii) 1 for Q_S, 1 for Q_C with sign, 1 for the new I_L. (iii) 1 for using 415 V and $Q_C/3$, 1 for C. (iv) 1 for the value with a valid reason.

COMMON MISTAKES:

- Using 18 kW as the motor input power. Always divide the shaft power by efficiency first (as in Jun23 Q4b).
- Adding power factors, or adding current magnitudes, instead of adding P and Q separately.
- Using 239.6 V for delta capacitors. That gives 311.8 µF, three times too big. 239.6 V is only correct if the capacitors are in star.
- Using Q_S (11.10 kVAr) as the capacitor size. The capacitor supplies the *difference*, $Q_T - Q_S$.

Question 2 marking guide

$A = 0.04 \times 0.03 = 12 \times 10^{-4}$ m², $\mu_0 = 4\pi \times 10^{-7}$ H/m.

(i)

$$\mathcal{R}_c = \frac{l_c}{\mu_0 \mu_r A} = \frac{0.36}{4\pi \times 10^{-7}(2500)(12 \times 10^{-4})} = \mathbf{95.49\ kA/Wb}$$

$$\mathcal{R}_g = \frac{g}{\mu_0 A} = \frac{0.4 \times 10^{-3}}{4\pi \times 10^{-7}(12 \times 10^{-4})} = \mathbf{265.3\ kA/Wb}\ \text{(each gap)}$$

The iron and the two gaps are in series:

$$\mathcal{R}_T = \mathcal{R}_c + 2\mathcal{R}_g = 95.49 + 2(265.3) = \mathbf{626.0\ kA/Wb}$$

$$L = \frac{N^2}{\mathcal{R}_T} = \frac{200^2}{626\,009} = \mathbf{63.90\ mH}$$

(ii) $\Phi = BA = 1.2 \times 12 \times 10^{-4} = 1.44$ mWb. From $NI = \Phi \mathcal{R}_T$:

$$I = \frac{\Phi \mathcal{R}_T}{N} = \frac{1.44 \times 10^{-3} \times 626\,009}{200} = \mathbf{4.507\ A}$$

Energy in the gaps (two gaps, each of volume Ag):

$$E_g = 2 \times \frac{B^2}{2\mu_0} \times Ag = 2 \times \frac{1.2^2}{2(4\pi \times 10^{-7})} \times (12 \times 10^{-4})(0.4 \times 10^{-3}) = \mathbf{0.5500\ J}$$

(Check: the total stored energy is $\frac{1}{2}LI^2 = 0.6490$ J. The gaps hold 85 % of it, which matches $2\mathcal{R}_g/\mathcal{R}_T$.)

(iii) Each gap has two facing surfaces of area A, and each gap pulls with

$$F_1 = \frac{B^2}{2\mu_0} A = \frac{1.2^2}{2(4\pi \times 10^{-7})}(12 \times 10^{-4}) = 687.5\ \text{N}$$

There are two gaps between the halves, so the total force is $F = 2F_1 = \mathbf{1375\ N}$.

MARKS: (i) 1 for $\mathcal{R}_c$, 1 for $\mathcal{R}_g$ and the total with **two** gaps, 1 for L. (ii) 1 for I, 1 for the gap energy. (iii) 1 for the formula per gap, 1 for doubling.

COMMON MISTAKES:

- Counting only one airgap. The question says the flux crosses two gaps in series.
- Using the formula-sheet force with $2A$ but then doubling again (4 times too big), or forgetting to double (half the answer). Think about it as “one force per gap, two

gaps".
- Using $\frac{1}{2}LI^2$ for "energy in the airgaps". That is the total energy, including the iron.

Turns ratio $a = 3300/240 = 13.75$, so $a^2 = 189.1$. Rated HV current $= 20\,000/3300 = 6.061$ A, so the SC test is at rated current.

OC test (LV side). The shunt branch is found on the side where the test was done, then referred to the HV side.

$$R_{C,LV} = \frac{V^2}{P} = \frac{240^2}{135} = 426.7\ \Omega$$

$S = VI = 240 \times 2.6 = 624$ VA, $\quad Q = \sqrt{S^2 - P^2} = \sqrt{624^2 - 135^2} = 609.2$ VAr, $\quad X_{M,LV} = \frac{V^2}{Q} = \frac{240^2}{609.2} = 94.55\ \Omega$

Refer to the HV side by multiplying by a^2:

$$R_C = 426.7 \times 189.1 = \mathbf{80.67\ k\Omega}, \qquad X_M = 94.55 \times 189.1 = \mathbf{17.88\ k\Omega}$$

SC test (HV side). These are already on the HV side.

$$R = \frac{P}{I^2} = \frac{290}{6.06^2} = \mathbf{7.897\ \Omega}, \qquad |Z| = \frac{V}{I} = \frac{145}{6.06} = 23.93\ \Omega$$

$$X = \sqrt{|Z|^2 - R^2} = \sqrt{23.93^2 - 7.897^2} = \mathbf{22.59\ \Omega}$$

MARKS: 1 each for R_C and X_M on the LV side, 1 for referring both to HV, 1 each for R and X.

COMMON MISTAKES:

- Forgetting that the OC test was done on the LV side, and quoting 426.7 Ω and 94.55 Ω as the HV values. They are 189 times too small.
- Referring the SC results as well. They were measured on the HV side already.
- $X_M = V/I$. That is the magnitude of the parallel pair, $|R_C \parallel jX_M|$ (92.3 Ω on the LV side), not X_M. Split P and Q first.

All on the HV side. $\mathbf{V}_2' = aV_2 = 13.75 \times 240 = 3300\angle 0^\circ$ V.

(i) Rated kVA at rated voltage means $I_2' = 20\,000/3300 = 6.061$ A. A leading load puts the current **ahead** of the voltage:

$$\mathbf{I}_2' = \mathbf{6.061\angle +36.87^\circ\ A} = 4.848 + j3.636\ \text{A}$$

KVL through the series branch:

$$\mathbf{V}_1 = \mathbf{V}_2' + \mathbf{I}_2'(R + jX) = 3300 + (4.848 + j3.636)(7.897 + j22.59)$$

$$= 3300 + (38.29 - 82.13) + j(109.5 + 28.71) = 3256.2 + j138.2 = \mathbf{3259\angle 2.43^\circ\ V}$$

(ii)

$$VR = 100\,\frac{|\mathbf{V}_1| - |\mathbf{V}_2'|}{|\mathbf{V}_2'|} = 100\,\frac{3259.1 - 3300}{3300} = \mathbf{-1.24\ \%}$$

It is negative because the leading current turns the $jX\mathbf{I}_2'$ drop so that it partly opposes $\mathbf{V}_2'$. The output voltage therefore rises above its no-load value as load is applied. (Check with the approximate formula $I_2'(R\cos\phi - X\sin\phi)/V_2' = -1.33$ %: same sign, similar size.)

(iii) $P_{out} = 20\,000 \times 0.8 = 16$ kW.

$$P_{CU} = I_2'^2 R = 6.061^2(7.897) = 290.1\ \text{W}, \qquad P_{FE} = \frac{|\mathbf{V}_1|^2}{R_C} = \frac{3259.1^2}{80\,667} = 131.7\ \text{W}$$

$$\eta = \frac{16\,000}{16\,000 + 290.1 + 131.7} = \mathbf{97.43\ \%}$$

(Using the 135 W OC-test value for the iron loss gives 97.41 %. Either gets the mark if the method is stated.)

(iv) Maximum efficiency is where copper loss equals iron loss. The full-load copper loss is 290.1 W and the iron loss at rated voltage is 135 W:

$$x = \sqrt{\frac{P_{FE}}{P_{CU,FL}}} = \sqrt{\frac{135}{290.1}} = 0.6822 \quad \Rightarrow \quad S = 0.6822 \times 20 = \mathbf{13.64\ kVA}$$

$$\eta_{max} = \frac{13\,644 \times 0.8}{13\,644 \times 0.8 + 2(135)} = \mathbf{97.59\ \%}$$

MARKS: (i) 1 for $\mathbf{I}_2'$ with the **+** angle, 1 for the phasor equation, 1 for $\mathbf{V}_1$ in polar form. (ii) 1 for the value with its sign, 1 for the reason. (iii) 1. (iv) 1 for the kVA, 1 for η_{max}.

COMMON MISTAKES:

- Writing $\mathbf{I}_2' = 6.061\angle -36.87^\circ$ for a leading load. That is the lagging answer: it gives $\mathbf{V}_1 = 3421\angle 1.35^\circ$ V and +3.68 % regulation.
- Adding magnitudes: $3300 + 6.061 \times 23.93$. This is a phasor sum. Go to rectangular form.
- Using I_1 (which includes the magnetising current) in the copper loss.
- Putting maximum efficiency at full load. It is where $P_{CU} = P_{FE}$.

Question 3 marking guide

(i) $n_S = 60f/p$ with p = pole-pairs. $60 \times 60/1164 = 3.09$, so p = 3 and $n_S = 3600/3 = 1200$ rpm. The machine has **6 poles**.

It is an **induction machine**, because 1164 rpm is not a synchronous speed for 60 Hz. It runs below $n_S = 1200$ rpm, with slip. A synchronous machine would run at exactly 1200 rpm.

(ii)

$$s = \frac{n_S - n}{n_S} = \frac{1200 - 1164}{1200} = \mathbf{0.03}, \qquad f_2 = sf = 0.03 \times 60 = \mathbf{1.8\ Hz}$$

(iii) Rotor copper loss is sP_{AG}, so the output can be at most $(1 - s)P_{AG}$:

$$\eta_{ideal} = 1 - s = \frac{n}{n_S} = \mathbf{97\ \%}$$

(iv) On the linear part of the curve, torque is proportional to the slip speed. Rated slip speed is $1200 - 1164 = 36$ rpm, so at half torque it is 18 rpm:

$$n = 1200 - 18 = \mathbf{1182\ rpm}$$

$$\frac{P}{P_{rated}} = \frac{\tau}{\tau_{rated}} \times \frac{n}{n_{rated}} = 0.5 \times \frac{1182}{1164} = \mathbf{50.77\ \%}$$

MARKS: (i) 1 for 6 poles, 1 for induction with the reason. (ii) 1 for both s and f_2. (iii) 1. (iv) 1 for the speed, 1 for the power.

COMMON MISTAKES:

- Using 50 Hz out of habit. This nameplate says 60 Hz.
- Answering "3 poles". p in $60f/p$ is pole-**pairs**.
- Saying the output power is exactly 50 %. The speed also rose a little.
- Halving the speed instead of halving the slip speed.

Work per phase of the **delta**: $V_1 = V_L$, $I_1 = I_L/\sqrt{3}$, $P_1 = P/3$.

Blocked-rotor test (series branch only):

$$V_1 = 92\ \text{V}, \quad I_1 = \frac{10}{\sqrt{3}} = 5.774\ \text{A}, \quad P_1 = \frac{620}{3} = 206.7\ \text{W}$$

$$R = R_1 + R_2' = \frac{P_1}{I_1^2} = \frac{206.7}{5.774^2} = 6.200\ \Omega \quad \Rightarrow \quad R_2' = 6.200 - 2.8 = \mathbf{3.400\ \Omega}$$

$$|Z| = \frac{V_1}{I_1} = \frac{92}{5.774} = 15.93\ \Omega, \qquad X = \sqrt{15.93^2 - 6.200^2} = \mathbf{14.68\ \Omega}$$

(If a split is needed, $X_1 = X_2' = 7.34\ \Omega$.)

No-load test (shunt branch only):

$$V_1 = 400\ \text{V}, \quad I_1 = \frac{4.2}{\sqrt{3}} = 2.425\ \text{A}, \quad P_1 = \frac{320}{3} = 106.7\ \text{W}$$

$$R_C = \frac{V_1^2}{P_1} = \frac{400^2}{106.7} = \mathbf{1500\ \Omega}$$

$S_1 = 400 \times 2.425 = 969.9\ \text{VA}$, $Q_1 = \sqrt{969.9^2 - 106.7^2} = 964.1\ \text{VAr}$, $X_M = \dfrac{400^2}{964.1} = \mathbf{166.0\ \Omega}$

(The no-load pf is only 0.11. The no-load current is almost all magnetising current. R_C here also includes friction and windage.)

MARKS: 1 for converting to delta phase quantities, 1 for R_2', 1 for X, 1 for R_C, 1 for X_M.

COMMON MISTAKES:

- Mixing star and delta: using star-equivalent test values ($V_1 = 92/\sqrt{3}$, $I_1 = 10$ A) with the delta winding's $R_1 = 2.8\ \Omega$. This gives $R = 2.07\ \Omega$, which is less than 2.8 Ω, so R_2' comes out negative. A negative resistance is a sure sign of a star/delta mix-up. (A consistent star-equivalent working, with $R_{1Y} = 2.8/3 = 0.933\ \Omega$, is valid: it gives $R_2' = 1.133\ \Omega$, $X = 4.893\ \Omega$, $R_C = 500\ \Omega$ and $X_M = 55.32\ \Omega$, exactly one-third of the delta values. But the question asks for per-phase-of-delta values.)
- Forgetting to subtract R_1 from the blocked-rotor R.
- $X_M = V_1/I_1$. Separate P and Q first.

(i) Six poles, so p = 3:

$$n_S = \frac{60 \times 50}{3} = \mathbf{1000\ rpm}, \qquad s = \frac{1000 - 970}{1000} = \mathbf{0.03}, \qquad \frac{R_2'}{s} = \frac{0.45}{0.03} = \mathbf{15.00\ \Omega}$$

(ii) Star, so $\mathbf{V}_1 = 415/\sqrt{3} = 239.6\angle 0^\circ$ V. Series branch: $X = X_1 + X_2' = 2.4\ \Omega$.

$$\mathbf{I}_2' = \frac{\mathbf{V}_1}{R_1 + R_2'/s + jX} = \frac{239.6}{15.6 + j2.4} = \mathbf{15.18\angle -8.75^\circ\ A}$$

Shunt branch at the terminals:

$$\mathbf{I}_C = \frac{239.6}{300} = 0.7987\ \text{A}, \qquad \mathbf{I}_M = \frac{239.6}{j40} = -j5.990\ \text{A}$$

$$\mathbf{I}_1 = \mathbf{I}_2' + \mathbf{I}_C + \mathbf{I}_M = (15.00 - j2.308) + 0.799 - j5.990 = 15.80 - j8.298 = \mathbf{17.85\angle -27.71^\circ\ A}$$

$$\text{pf} = \cos 27.71^\circ = \mathbf{0.8854\ lagging}$$

(iii)

$$P_{IN} = 3V_1 I_1 \cos\phi = 3(239.6)(17.85)(0.8854) = \mathbf{11.36\ kW}$$

$$P_{AG} = 3I_2'^2 \frac{R_2'}{s} = 3(15.18)^2(15) = \mathbf{10.37\ kW}$$

$$\omega_S = \frac{2\pi \times 1000}{60} = 104.7\ \text{rad/s}, \qquad \tau = \frac{P_{AG}}{\omega_S} = \frac{10\,370}{104.7} = \mathbf{99.03\ Nm}$$

(The formula-sheet torque formula gives the same 99.03 Nm.)

$$P_{OUT} = (1 - s)P_{AG} = 0.97 \times 10\,370 = 10.06\ \text{kW}, \qquad \eta = \frac{10.06}{11.36} = \mathbf{88.56\ \%}$$

Loss check: stator Cu $= 3(15.18)^2(0.6) = 414.8$ W, iron $= 3(239.6)^2/300 = 574.1$ W, rotor Cu $= sP_{AG} = 311.1$ W. $10\,059 + 414.8 + 574.1 + 311.1 = 11\,359$ W $= P_{IN}$.

(iv)

$$s_{\tau\max} = \frac{R_2'}{|R_1 + jX|} = \frac{0.45}{\sqrt{0.6^2 + 2.4^2}} = \mathbf{0.1819}, \qquad n = (1 - 0.1819)(1000) = \mathbf{818.1\ rpm}$$

(Not asked, but useful: $\tau_{max} = 267.5$ Nm, 2.7 times the torque at 970 rpm.)

MARKS: (i) 1 for n_S, 1 for s and R_2'/s. (ii) 1 for $\mathbf{I}_2'$, 1 for adding the shunt currents to get $\mathbf{I}_1$, 1 for pf. (iii) 1 for P_{IN}, 1 for P_{AG} and τ, 1 for η. (iv) 1 for both.

COMMON MISTAKES:

- Using 415 V as the phase voltage of a star machine. Torque comes out 3 times too big.
- Using the rotor speed ω with P_{AG}. P_{AG} goes with ω_S. P_{OUT} goes with ω.
- Forgetting the shunt branch in $\mathbf{I}_1$. It does not change the torque, but it does change I_1, the pf and P_{IN}.
- In (iv), using $R_1 + X$ or X alone instead of the magnitude $|R_1 + jX|$.

Question 4 marking guide

(i)

$$Z_B = \frac{V_B^2}{S_B} = \frac{(13.8\times10^3)^2}{60\times10^6} = \mathbf{3.174\ \Omega}, \qquad I_B = \frac{S_B}{\sqrt{3}V_B} = \frac{60\times10^6}{\sqrt{3}\times 13\,800} = \mathbf{2510\ A}$$

(ii) At the same field current, the OC phase voltage and the SC current give the synchronous impedance:

$$|Z_S| = \frac{V_{OC}/\sqrt{3}}{I_{SC}} = \frac{9200/\sqrt{3}}{1250} = 4.249\ \Omega$$

$$X_S = \sqrt{|Z_S|^2 - R^2} = \sqrt{4.249^2 - 0.05^2} = \mathbf{4.249\ \Omega}, \qquad X_S = \frac{4.249}{3.174} = \mathbf{1.339\ pu}$$

(R is only 0.0158 pu, so it barely changes X_S.)

(iii) OC voltage is proportional to field current on the linear part:

$$I_F = 180 \times \frac{13.8}{9.2} = \mathbf{270\ A}$$

MARKS: (i) 1 for both bases. (ii) 1 for $|Z_S|$ with the $\sqrt{3}$, 1 for X_S in Ω, 1 for pu. (iii) 1.

COMMON MISTAKES:

- $X_S = 9200/1250 = 7.36\ \Omega$. The OC voltage is a line value. The circuit is per phase.
- Using $Z_B = V_B^2/(3S_B)$ or V_B/I_B with the line voltage. With line voltage and three-phase power, $Z_B = V_B^2/S_B$.

(i) $\mathbf{V} = 1\angle 0^\circ$ pu.

$$S = \frac{45}{0.85} = 52.94\ \text{MVA} = 0.8824\ \text{pu}$$

The generator delivers at a leading pf, so (generator convention, current out of the machine) the current leads the voltage:

$$\mathbf{I} = \frac{S}{V}\angle +\cos^{-1} 0.85 = \mathbf{0.8824\angle +31.79^\circ\ pu} = 0.75 + j0.4648\ \text{pu}$$

$$I = 0.8824 \times 2510 = \mathbf{2.215\ kA}$$

(ii) Generator: $\mathbf{E}_0 = \mathbf{V} + jX_S\mathbf{I}$ (R neglected).

$\mathbf{E}_0 = 1 + j1.339(0.75 + j0.4648) = 1 - 0.6224 + j1.004 = 0.3778 + j1.004 = \mathbf{1.073\angle 69.38^\circ\ pu}$

Rotor angle $\delta = \mathbf{69.38}^\circ$.

Q4b phasor diagram: I = 0.8824 pu leading by 31.79 degrees, E0 = 1.073 at 69.38 degrees

Field current, using the excitation scale from 4a(iii) (270 A per pu of E_0):

$$I_F = 1.073 \times 270 = \mathbf{289.6\ A}$$

Check: $P = E_0 V \sin\delta / X_S = 1.073 \times \sin 69.38^\circ / 1.339 = 0.75$ pu = 45 MW.

(iii) With E_0 and V fixed, the maximum power is at $\delta = 90^\circ$:

$$P_{max} = \frac{E_0 V}{X_S} = \frac{1.073 \times 1}{1.339} = 0.8013\ \text{pu} = \mathbf{48.08\ MW}$$

60 MW is not possible. The generator would lose synchronism (pole slip) before reaching it. To make 60 MW (1 pu) even theoretically possible, $E_0 \geq X_S \times 1/V = 1.339$ pu, so

$$I_F \geq 1.339 \times 270 = \mathbf{361.4\ A}$$

(In practice you would want even more field current, so that δ stays well below 90°.)

MARKS: (i) 1 for $\mathbf{I}$ with the + angle, 1 for the kA value. (ii) 1 for the phasor equation, 1 for E_0 and δ, 1 for I_F. (iii) 1 for P_{max} in MW with the conclusion, 1 for the minimum field current.

COMMON MISTAKES:

- $\mathbf{I} = 0.8824\angle -31.79^\circ$. That is lagging. It gives $E_0 = 1.908\angle 31.75^\circ$ pu, a different machine state.
- Using $\mathbf{E}_0 = \mathbf{V} - jX_S\mathbf{I}$. That is the motor equation with input current. This is a generator, and the course's circuit has the current flowing out of $\mathbf{E}_0$.
- Assuming that an under-excited (leading pf) generator must have $E_0 < 1$. Here $E_0 = 1.073$ pu. "Under-excited" means $E_0 \cos\delta < V$ (here 0.378 < 1), so the machine absorbs reactive power.
- In (iii), using $E_0 = 1$ pu or forgetting to convert the pu answer to MW.

You do not need every point for full marks. Roughly one mark per well-explained point.

(i) Over- and under-excitation [3 marks]

- The field current sets the excitation voltage. When a generator is connected to a large grid, the grid fixes the terminal voltage and the frequency.
- **Over-excited:** the field current is higher than needed for the terminal voltage. The generator delivers reactive power to the grid and runs at a lagging pf, like a capacitor seen from the grid.
- **Under-excited:** the field current is lower. The generator absorbs reactive power from the grid and runs at a leading pf. This weakens the magnetic coupling, so the maximum power before losing synchronism is lower (as in 4b(iii)).
- **Mechanical input power (prime mover) controls real power:** more input torque advances the rotor angle, and more real power is delivered. The speed stays at synchronous speed.
- **Field current controls reactive power:** at a fixed real power, changing the field current mainly changes the reactive power and the pf.

(ii) Rotor resistance vs V/f speed control [3 marks]

- In an induction machine, the airgap power splits into rotor copper loss (the slip fraction) and mechanical power (the rest). So the ideal efficiency is one minus the slip, the ratio of rotor speed to synchronous speed.
- Extra rotor resistance lowers the speed by increasing the slip, while the synchronous speed stays the same. At large slip, a large part of the airgap power is wasted as heat in the rotor circuit and the external resistance. At half speed, at least half the airgap power is lost.
- An inverter that lowers the frequency lowers the synchronous speed itself. The torque–speed curve slides to lower speed with the same shape, so the motor still runs at a small slip and the rotor losses stay small. Efficiency stays high.
- The voltage is reduced in proportion to the frequency to keep the flux constant. This avoids saturation and keeps the maximum torque available.

(iii) Per-unit system [2 marks]

- Quantities are expressed as fractions of chosen base values, so the normal operating values are near 1 and abnormal values are easy to see.
- If the base voltages on each side of a transformer are chosen in the ratio of its turns, the per-unit impedance is the same from both sides, and an ideal transformer disappears from the per-unit circuit. You do not have to refer impedances across each transformer.
- Machine and transformer parameters fall in narrow per-unit ranges regardless of size, so values are easy to check and compare. The single-phase and three-phase per-unit values are the same, which removes the factors of 3 and root 3.

> **COMMON MISTAKES:**

- Using equations or symbols. The question says words only.
- In (i), mixing up which knob controls what. Field current is for reactive power. Prime mover is for real power.
- In (ii), saying the first method is inefficient “because of the resistor” without saying why: the slip is large and rotor loss is the slip fraction of the airgap power.

Score interpretation

Mark each part strictly, then add up each question out of 20. Use the table below to decide what to do in your final 24 hours. Do not try to re-learn everything. Fix the one or two weakest areas, and then do the “everyone” list.

Your result	What it means	What to do
64–80 (80 %+)	Exam-ready.	Light review only. Re-do one part you lost marks on, then read through your reference sheet and the 4c model points. Sleep.
48–63 (60–79 %)	Solid. One or two topics are leaking marks.	Use the per-question rows below for any question under 15/20. Spend about 60 minutes on each weak question, not more.
32–47 (40–59 %)	Method is there but errors cost a lot.	Look at *why* you lost marks. If it was mostly signs, $\sqrt{3}$ and star/delta, re-do the traps sections of M2, M7 and M10. If it was not knowing where to start, work through the playbooks (section 2) of the packs for your two lowest questions.
Below 32	Big gaps.	Pick the two questions where you scored lowest, and do only those packs' method and worked examples. A clean 15/20 on two questions is worth more than 10/20 on four.

Per question, if you scored under 15/20:

Lost marks in	Most likely cause	Go to
Q1a, 1b	Power flow, delta vs star, $\sqrt{3}$ in the wrong place	**M1** (power, phasors, complex power), **M2** section on star/delta
Q1c	PFC: forgetting efficiency, wrong sign of Q_C, delta capacitor voltage	**M2** (PFC playbook), then redo Jun24 Q1d and Jun23 Q4b in **MX1/MX2**
Q2a	Reluctance in series, gap counting, energy and force	**M3**
Q2b	Test data: which side, referring with a^2, splitting P and Q	**M4**
Q2c	Leading vs lagging $\mathbf{I}_2'$, phasor sum, regulation and efficiency	**M5**
Q3a	Poles and n_S, slip, ideal efficiency, linear τ–n curve	**M6**
Q3b	No-load and blocked-rotor tests, per-phase delta conversion	**M8** (section 1.2), then M2's equivalent-circuit tests
Q3c	Equivalent-circuit solve, power flow, torque, $s_{\tau\max}$	**M7** (method), then **M9** (exam-style problems)
Q4a	X_S from OC/SC tests, base values	**M10**
Q4b	Generator convention, leading current, E_0, δ, P_{max}	**M11**
Q4c	Short answers too vague, or with equations	Re-read the short-answer sections in **M5**, **M8** (section 1.8) and **M11**, and the official 4c model points in **MX1**/**MX2**. Practise writing 3 bullet points per question in 4 minutes.

Everyone, in the final 24 hours (about 2 hours in total):

1. Re-do, without looking, every part where you lost marks in this mock (30 min).
2. Check your printed reference sheet has: star/delta rules, the PFC steps, $\mathcal{R} = l/(\mu_0 \mu_r A)$, the OC/SC test formulas, the transformer phasor equation, the IM simplified circuit and power flow, $s_{\tau\,\max}$, the OC/SC X_S method and $\mathbf{E}_0 = \mathbf{V} + jX_S\mathbf{I}$ (15 min).
3. Write a one-line "sign check" on the sheet: lagging current is $\angle -\phi$, leading is $\angle +\phi$; generator slip is negative; generator δ is positive (5 min).
4. Do the **R2** quick recall questions on Ch7 and Ch8 (20 min), since these two chapters were not in the mid-semester test.
5. Read the 4c model points from Jun23, Jun24 and this mock out loud once (10 min). Then stop and rest.